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Fine-Tuned Generative AI for Automated Structured Data Extraction and Insight Generation from Legacy Petroleum Well Reports: An Egyptian Oilfields Case Study

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Abstract

This paper introduces a workflow based on fine-tuned generative AI to automate the extraction of structured data and insights from legacy petroleum well reports. It replaces tedious manual analysis of documents (e.g., Daily Drilling Reports and Workover Reports) with an automated process that loops through the data and outputs a structured, queryable dataset. The scope includes diverse operational reports from Egyptian oilfields, demonstrating practical applications for improved resource management.

The approach consists of three main stages. First, operational reports were collected from various Egyptian oil fields in formats such as PDF, Excel, and Word. Second, a foundational large language model was used to parse and interpret the content, producing a clean, domain-specific, labeled dataset. Third, this dataset was used to fine-tune compact generative AI models (1.5B–7B parameters) for local deployment. These models were trained to perform tasks such as jargon translation, report summarization, title generation, and extracting key information on well history, operational problems, and their solutions, enabling structured, consistent datasets and actionable insights.

An initial assessment by domain experts indicates that the fine-tuned local model (7B parameters) achieves reliable results. The performance and validation were based mainly on real operational reports, with the extracted and processed information verified by domain experts, providing more reliable validation than evaluation metrics. The AI-driven approach significantly reduces processing time compared to manual analysis while maintaining consistency in the extracted structured data, making it suitable for our main objective that is automation. Its capacity to understand technical jargon and generalize across various report formats depends heavily on the quality of the fine-tuning dataset. Achieving these results with the fine-tuned 7B-parameter model suggests strong potential for even better performance and generalization by tuning larger models and collecting more high-quality, informative datasets.

This research presents a novel two-stage AI approach for the energy sector. First, large language models extract and structure a domain-specific, curated dataset from complex well reports. Then, this dataset is used to fine-tune a smaller model optimized for fast, local deployment. The workflow offers a scalable solution for building specialized AI tools that deliver specialized, accurate results. It marks a forward step in applying generative AI and LLMs to petroleum data analytics.

Introduction

The natural data-intensive nature of the global energy sector, and in particular the oil & gas industry generates huge amounts of associated information: at exploration, drilling, production, and even maintenance procedures. This wide but condensed data livestock is key to decision-making, operational efficiency and sustainability. However, as much as 80% of this valuable information exists in unstructured formats — like daily drilling reports (DDRs), workover reports (WORs), presentations, spreadsheets and historical well logs. However, despite being mostly unstructured and stored in pdfs/excel/word documents (legacy formats), there is a treasure trove of data available to derive meaningful operational procedures as well as historical insights from the production — largely an untapped source for analysis ([Iraya Energies, 2021](#)).

Artificial Intelligence (AI) and Machine Learning (ML) technologies have become revolutionary tools in several fields, including oil and gas, providing powerful solutions for intricate data issues ([Iraya Energies, 2021](#)). Even the Society of Petroleum Engineers (SPE) acknowledged such potential, having a special branch for "artificial intelligence in prediction analysis" since 2009 to facilitate AI technology in the industry ([Al-Quraishi and Al-Ameri, 2024](#)). More recently, Large Language Models (LLMs) with their potential to process and generate human-like text have transformed Natural Language Processing (NLP) tasks (Repsol, n.d.). Their ability to deal with enormous volumes of unstructured textual data and still identify meaningful patterns makes them strongly suitable for solving the specific data issues in the energy industry ([Al-Ameri, 2022](#)).

In spite of sustained efforts towards the digitalization of the oil and gas sector, the inefficient exploitation of unstructured data continuously prevented the success of such efforts. This situation presents a strategic hindrance that hinders organizations from realizing projected gains, which can include a possible sector value of \$1 trillion USD, 10% production cost savings, as well as a 10% increase of the success rate of resource exploitation ([Iraya Energies, 2021](#)). The challenge goes beyond data storage, involving substantial issues of accessibility as well as interpretability, such that relevant historical data become ever more evasive, defying evaluation as well as integration efforts, ultimately leading to a significant lack of knowledge within organizations ([Srivastava et al., 2021](#)). The current research presented here directly addresses such a hindrance through an attempt to transform hitherto considered "unused" data into a strategic asset, ultimately supporting the holistic realization of digital transformation across the entire energy sector ([Iraya Energies, 2021](#)).

This research introduces an original workflow that employs fine-tuned generative AI to automate the extraction of structured data as well as the production of insights from voluminous legacy well reports of the petroleum sector. It aims at transforming the historically inefficient manual processing of unstructured Daily Drilling Reports (DDR) and Workover Reports (WOR) through the application of an efficient, AI-powered process. Through processing such as jargon translation, summary, title identification, as well as the detection of well history, issues, as well as fixes, the new methodology aspires to link raw data to actionable knowledge, hence fostering knowledge discovery as well as democratization of same across the company. Focusing on various operational data drawn from numerous Egyptian oilfields, the study reveals a real-world application towards enhanced regional resource optimization, hence contributing significantly to the digitalization of the oil and natural gas sector ([Iraya Energies, 2021](#)).

Problem Definition

The oil sector faces an omnipresent challenge related to effectively leveraging its large data landscape, of which about 80% of the data is found to be unstructured ([Iraya Energies, 2021](#)). It includes vital operating documentation such as Daily Drilling Reports (DDRs), Workover Reports (WORs), well logging, seismic data, as well as maintenance reports ([SPE, 2023](#)). These reports are often provided in archaic formats (like PDFs, scanned documents, handwritten documents, etc.) that do not adhere to agreed-on schemas, hence

causing difficulty when processed through standard database systems, hence triggering significant manual intervention for effective administration (Grooper, n.d; [Iraya Energies, 2021](#)).

Scope of the Problem. The prevailing reliance on manual methods for analyzing these unstructured reports is highly inefficient, time-consuming, and prone to human error ([USGS, 2016](#)). Engineers and geoscientists reportedly spend 50-80% of their time searching for and assembling data rather than performing higher-value analytical tasks ([Iraya Energies, 2021](#)). This translates into significant operational costs and delays in critical decision-making (Grooper, n.d.). For instance, manual screening of DDRs is a "tedious time-consuming exercise" (Dataiku, n.d.), and processing drilling mud reports is "labor-intensive, costly, and error-prone" ([Pallanich, 2025](#)). The lack of reporting standards, prevalent use of niche technical abbreviations, and highly tabular document structures further complicate manual extraction, contributing to data inconsistencies ([USGS, 2016](#)).

Consequences of inefficient data handling are severely negative on decision-making activities, operational performance, as well as resource deployment. Inability to have effective access to or make use of unstructured historical data leads to "ill-informed investment decisions" and may significantly impact "operational planning and decision-making". As a result, such situations create operational inefficiencies, budget shortages, as well as suboptimal use of resources across the value chain. In industrial environments, situations are frequently described under terms of, "I don't know what I don't know," and "if we only knew what we know," evidencing a significant concern about accessibility as well as utilization of knowledge ([Iraya Energies, 2021](#)). Additionally, data on production volumes, usually acquired through manual means, may be "incorrect or inconsistent," hence, stringent quality assurance procedures may be required ([Imperial College London, 2020](#)).

Rationale for an AI-Based Solution. At its core, the challenge goes beyond incremental improvement of the productivity of data analysis; it addresses the significant value lost as a consequence of lost opportunities and suboptimal strategic choices ([Iraya Energies, 2021](#); Dataiku, n.d.). The transformation of hitherto unrelated, unusable data into a strategically significant asset shifts the focus from rudimentary cost reductions to real value creation. Furthermore, the intrinsic complexity of the data poses a distinctive challenge: standard petroleum well reports exhibit significant variability of data, which manifests as "dissimilar attribute representations" as well as "heterogeneous document formats" ([SPE, 2023](#)). The presence of "specialized, technical abbreviations" as well as extremely structured tabular formats renders automated processing even more difficult ([USGS, 2016](#)). This demands the application of state-of-the-art AI models that have the ability to effectively cope with such variability, noise, as well as domain-specific vocabulary.

Rationale for Our Research. This research addresses head-on the identified challenges through the deployment of an original workflow that employs optimally tuned generative artificial intelligence to enable automated structured data extraction and insight generation from extensive legacy petroleum well reports. The overarching aim is to optimize the presently inefficient manual analysis of unstructured Daily Drilling Reports (DDR) and Workover Reports (WOR), which cover a broad spectrum of operational data from various Egyptian oilfields ([Iraya Energies, 2021](#)). Our methodology embraces a revolutionary two-phase artificial intelligence system: (1) application of a state-of-the-art foundation large language model to parse raw legacy documents (PDFs, Excel, Word), hence producing a high-quality, domain-specific dataset of carefully cleaned well data with accurate interpretation; and (2) subsequently, exploiting the fine-tuned dataset to optimize smaller, lightweight generative AI models (e.g., covering 1.5 to 7 billion parameters or even smaller) for localized, pointwise deployment. Optimized models executing tasks such as jargon interpretation, summary, extraction of operational data, title generating, as well as identification of issues and resolution, finally produce structured insights.

Prior Research. The application of Artificial Intelligence (AI) in petroleum engineering dates back to the 1970s, with the Society of Petroleum Engineers (SPE) formally recognizing AI in "prediction analysis" in 2009 (Al-Quraishi and Al-Ameri, 2024). Early AI and Machine Learning (ML) applications focused on optimizing oilfield development, including production prediction, development plan optimization, residual oil identification, fracture identification, and enhanced oil recovery (Al-Quraishi and Al-Ameri, 2024). ML algorithms were successfully deployed for tasks like drill bit diagnosis, well-log analysis, drilling surveillance, reservoir simulation (seismic pattern identification, history matching, reservoir characterization), and petrophysical property prediction (permeability, porosity). These applications demonstrated AI's potential to handle complex, large-scale data, often outperforming traditional geostatistical approaches in cost, precision, and speed (Al-Mudhafar, W, 2022).

In spite of such evolutions, the sector has struggled to economically deal with large volumes of unstructured data, while becoming expert at structured data management. Over the last decade, projects involving the automatic extraction of data from drilling reports have emerged, employing Machine Learning techniques like Natural Language Processing (NLP) for textual data as well as Deep Convolutional Neural Networks for non-text data (Iraya Energies, 2021). These techniques have effectively detected trouble-prone wells, classified drilling incidents, as well as extracted specific attributes (casing size, depth, well ID, ROP) from Daily Drilling Reports (DDRs). Significantly, however, challenges abound, including robustness to OCR errors (stemming from poor print quality or illegible handwriting), adaptability to deal with the diversity of document formats, as well as efficient multi-source data integration. Many existing solutions, too, require significant customization when faced with new document structure, which limits scalability as well as generality (SPE, 2017 & 2023).

The development of Large Language Models (LLMs), including OpenAI's GPT-3, Google's BERT, and the Llama series, which have billions of parameters, has significantly improved Natural Language Processing (NLP)'s capabilities (Ferrigno et al., 2024). Trained on large, diversified text data, such base models show unprecedented knowledge of various styles of language, cultural nuances, as well as subject matter. Advanced text generation, translation, summarization, sentiment analysis, as well as question answering, form their all-embracing capabilities. For the oil and gas industry, LLMs are increasingly explored for value addition through automating tasks, enhanced predictive ability, intelligent decision-making, as well as workflow optimization through processing of unstructured textual data from reports, logs, as well as seismic data. Firms such as Repsol as well as Petrobras employ LLMs to optimize operational efficiency, streamline decision-making, augment communication, simplify workflow, as well as report summarization (Repsol, n.d.).

The progression from traditional AI/ML to generative AI, particularly LLMs, marks a qualitative shift in unstructured data processing within the oil and gas industry. While traditional ML excelled at classification, prediction, or predefined attribute extraction (Al-Mudhafar, W, 2022), LLMs "revolutionize natural language processing tasks" (Ferrigno et al., 2024) and can "provide solutions and create relevant content" (Repsol, n.d.). This distinction is crucial as LLMs move beyond merely analyzing existing data to actively generating new insights and interpreting complex narratives. This capability is vital for tasks like "jargon translation" and "identifying well history, problems, and solutions" from free-text reports, requiring nuanced contextual understanding rather than just discrete factual extraction. This highlights that current research represents a foundational transformation in leveraging unstructured data.

Moreover, there is a significant lack of literature on the overall synthesis of multi-faceted insights from the entirety of complicated, unstructured well heritage documentation. Unlike previous automated extraction attempts from drilling reports (USGS, 2016), which targeted minimal, pre-defined data elements (such as casing size, rate of penetration, and identification of trouble well), the current research addresses a more substantial part of the broader concept of "insight generation".

The abstract of this work specifically describes tasks like "jargon translation, summarization, building of titles, and identification of well history, issues, and resolution," which involve a deeper, narrative-driven

understanding and synthesis of the information. This process goes beyond identification of facts, seeking to provide a more substantive operational summary aiding effective asset handling as well as long-range planning of a strategic kind.

Literature Review

The oil and gas industry has a significant data dependence, as well as significant volumes of data generated during the process of exploration, drilling, production. This data plays a critical role in decision-making, optimization of operations, as well as resource planning. Nevertheless, about 80% of such meaningful data remain in unstructured formats, including spreadsheets, daily drilling reports (DDR), as well as well logs. Although such documents contain vital knowledge, they remain underutilized because of the challenges faced when processing as well as analyzing them (Wu et al., 2025).

Manually analyzing these unstructured reports is inefficient, slow, and prone to error. Engineers spend 50-80% of their time finding data, not analyzing it, which causes high costs and delays. For example, screening drilling reports is "tedious," and processing mud reports is "labor-intensive, costly, and error-prone". This inability to use historical data leads to "ill-informed investment decisions" and a "knowledge gap". This prevents the industry from achieving its full potential, including a projected trillion-dollar value and a 10% increase in recovery rates (Wu et al., 2025).

AI and ML are Vital for managing data in the oil and gas industry. The Society of Petroleum Engineers (SPE) has long explored this field (Nadaş et al., 2025). The recent rise of Large Language Models (LLMs) has advanced Natural Language Processing (NLP) by enabling the interpretation of vast, unstructured text for tasks such as "jargon translation" and "identifying well history" (Johnson & Hyland-Wood, 2025; Naveed et al., 2023; Wu et al., 2025). This work introduces a two-phase methodology that applies generative AI to automate data extraction and generate insights from historical petroleum reports. The work merges existing knowledge, identifies trends, and highlights gaps, while also evaluating the methodology's effectiveness and challenges within the broader landscape of AI in the oil and gas sector.

Evolution of AI and Natural Language Processing in Petroleum Engineering

Early Applications of AI/ML in Oilfield Development (1970s–2010s). AI has been a part of petroleum engineering since the 1970s, with the Society of Petroleum Engineers (SPE) officially recognizing its role in "prediction analysis" by 2009. Early AI and ML applications focused on optimizing specific tasks with structured data, such as production prediction, well-log analysis, and enhanced oil recovery (Li et al., 2020). However, the industry struggled with the huge volume of unstructured data. Modern Large Language Models (LLMs) are now addressing this by democratizing knowledge, a key shift for the field (Salem et al., 2022).

Natural Language Processing Innovations for Information Extraction from Perforation Data. In the past decade, studies have focused on using Machine Learning and Natural Language Processing (NLP) for automated data extraction from drilling reports (Srivastava et al., 2021). While these approaches have shown success in identifying defective wells and attributes like Rate of Penetration (ROP), they face significant challenges, including Optical Character Recognition (OCR) errors and a lack of flexibility with disparate document formats. Crucially, prior methods extracted only pre-defined data points and failed to provide a comprehensive, narrative understanding, a deficiency that is now being addressed by Large Language Model refinement.

Important Transformations Relating to Large Language Models (LLMs). Large Language Models (LLMs) like GPT-3, BERT, and Llama have transformed Natural Language Processing (NLP) with their ability to process vast and varied datasets (Naveed et al., 2023; Johnson & Hyland-Wood, 2025). In the oil and gas industry, companies are using LLMs to automate processes and inform decisions by processing

unstructured reports and logs (Povoas et al., 2025). For example, Petrobras uses Azure OpenAI Service to streamline workflows for its 110,000 employees, and Aramco is developing an energy-specific LLM to help professionals quickly gain insights (Povoas et al., 2025; Pallanich, 2025). This shift from traditional AI to generative AI is significant because LLMs can do more than just classify data. They can generate new content, provide solutions, and interpret complex narratives to perform tasks like "jargon translation" and identify well history or issues from free-text reports (Povoas et al., 2025). (Table 1) summarizes this evolution of AI and ML in the industry.

Table 1—Evolution and Applications of AI/ML/NLP in Oil & Gas Data Management

Technology/Method	Key Application Areas	Key Contributions/Capabilities	Limitations/Challenges	Reference
Traditional AI/ML (Pre-LLM)	Production Prediction, Reservoir Characterization, Well-Log Analysis, Seismic Interpretation, Drilling Optimization, Fracture Identification, Enhanced Oil Recovery, Drill Bit Diagnosis	Optimized drilling, improved seismic interpretation, automated data compilation, risk management, petrophysical property prediction (permeability, porosity). Often outperformed traditional geostatistical approaches in cost, precision, and speed.	High input parameter requirements, lower accuracy for some algorithms, sensitivity to data quality, limited to structured data management, struggled with unstructured data.	(Li et al., 2020)
Early Automated Information Extraction (NLP/DNN)	Specific Attribute Extraction from DDRs/WORs (e.g., casing size, depth, well ID, ROP), Problematic Well Identification, Drilling Dysfunction Classification	Automated data compilation for specific fields, some noise reduction, can leverage pre-trained models with less training data.	Customization needed for diverse documents, robustness against OCR inaccuracies (poor print/handwriting), adaptability to diverse document layouts, limited to specific facts rather than holistic insights, scalability and generalizability issues.	(Srivastava et al., 2021)
Foundational Large Language Models (LLMs)	General Text Generation, Translation, Summarization, Question Answering, Sentiment Analysis, Streamlining Workflows, Automating Processes, Enhancing Predictive Capabilities	Handling massive unstructured text, generating human-like content, broad applicability, emergent abilities (reasoning, planning, decision-making), can understand various language styles and cultural contexts.	General models lack domain specificity, potential for imprecise information in niche areas (hallucinations), high computational cost for training and inference, extensive hardware requirements, data privacy concerns with proprietary models.	(Povoas et al., 2025)
Fine-tuned LLMs (Proposed & Emerging)	Domain-Specific Data Extraction, Jargon Translation, Holistic Insight Generation, Local Deployment, Predictive Maintenance, Drilling Optimization, Reservoir Management, Knowledge Management, Automated Document Generation	High accuracy in specialized domains, contextual understanding, cost-efficiency, data security (local deployment), rapid deployability, addresses data scarcity through synthetic data generation, improved understanding of specialized terminology and nuances, can cite original sources (with RAG).	Requires high-quality domain-specific data for training (though synthetic data helps), computational intensity of fine-tuning (though PEFT helps), still faces challenges with interpretability and ensuring factual accuracy (hallucinations).	(SPE JPT, 2025)

Significance of Innovations and Latest Techniques for Comprehending Petroleum Sciences

Relevance Optimization for the Domain. While base Large Language Models (LLMs) have exceptional versatility, their generalist nature can limit their accuracy in specialized fields like petroleum engineering, where unique vocabulary and context can lead to the proliferation of false information (Naveed et al., 2023). To address this, fine-tuning adjusts a pre-trained LLM using a domain-specific dataset, enabling it to understand specialized jargon and reduce errors (Durai, 2025; Moonlight Review, 2024; Wu et al., 2025). The quality of this data is a key requirement for success (Wu et al., 2025). For instance, the Energy LLM (ELLM) project fine-tuned the Llama 3 model on data from the OnePetro database, creating a specialized tool for the oil and gas industry (Pallanich, 2025).

Handling Sparsity of Data through Synthetic Data Generation. A major challenge in domain-specific AI is the lack of high-quality, labeled training data. Manually labeling large amounts of specialized data is difficult and expensive, which has limited AI adoption (Wu et al., 2025). However, generative AI, specifically Large Language Models (LLMs), has a solution in synthetic data generation (Zhu et al., 2025). LLMs can create high-quality, coherent text that closely mimics human-written content, effectively augmenting or replacing real datasets when labeled data is scarce or costly. For example, GPT-3 can generate thousands of examples for sentiment analysis in minutes at a fraction of the cost of human labeling. This ability to produce synthetic data at a massive scale is crucial for training data-hungry models and creating balanced datasets (Nadaş et al., 2025).

The Structure of Dual-Phase Generative AI. This work introduces a new two-phase AI system that first uses a cornerstone Large Language Model (LLM) to generate a high-quality, synthetic dataset from historical documents, effectively overcoming the data annotation bottleneck. This data is then used to fine-tune compact generative AI models (1.5B to 7B parameters) for specialized, on-site deployment. This "pragmatic AI" approach is a significant shift away from sole reliance on costly, cloud-based LLMs (Naveed et al., 2023), prioritizing cost-effectiveness, data security, and real-time processing (Povoas et al., 2025). The methodology aims to provide deep, narrative-based insights—such as jargon translation, summarization, and well history—from unstructured reports. Its successful application to varied datasets from Egyptian oilfields demonstrates its robust versatility and real-world feasibility.

Varied Perspectives and Ongoing Discussion

Level of Intensity of Hallucinations. LLMs face a major concern known as hallucinations, where they produce confident but false information, including fabricated facts or citations (Johnson & Hyland-Wood, 2025; Tariq, 2025). This is a significant risk in the oil and gas industry, where such errors could be costly and dangerous, and are considered unacceptable (Low, 2025).

To mitigate this, strategies like Retrieval-Augmented Generation (RAG) allow LLMs to access external knowledge bases to improve accuracy (Pallanich, 2025). Other methods include careful prompt design, human oversight, and fine-tuning on verified data. Although state-of-the-art models have reduced hallucination rates to below 2% by 2025, continued research and development in this area remain critical (Tariq, 2025).

What is Interpretability. Large language models often act as "black boxes," making it difficult to understand how they reach a decision (Linardatos et al., 2020). This lack of transparency, known as the interpretability problem, hinders their use in vital fields like petroleum engineering, where understanding a model's prediction is essential for building trust and ensuring accountability (SPE JPT, 2025). While fine-tuning can increase some interpretability, the field of Explainable AI (XAI) is actively working to balance a model's performance with its transparency (Linardatos et al., 2020; Wu et al., 2025).

Protecting Data Privacy and Security. The extensive data needed to train LLMs raises significant concerns about data privacy and security (Johnson & Hyland-Wood, 2025). There is a risk of sensitive information leaking through memorization or inference attacks. This requires robust AI data governance frameworks to ensure data is used responsibly, confidentially, and ethically. Effective governance is crucial for minimizing breaches, enhancing security, and ensuring regulatory compliance (SPE JPT, 2025). The challenge is made worse by the rapid increase in data volume and the need for secure integration from multiple sources (Pahune et al., 2025).

Identification of Gaps of Existing Literature and Novel Contributions of the Present Research

Overcoming prior limitations in AI, this work presents a two-phase methodology for full-spectrum automation (Wu et al., 2025). Earlier AI and ML applications succeeded with structured data but failed to process unstructured legacy reports. While Natural Language Processing (NLP) later extracted specific attributes, it was hindered by Optical Character Recognition (OCR) errors and a lack of scalability (Wu et al., 2025). Furthermore, core Large Language Models (LLMs) proved to be too generic for specialized fields like petroleum engineering, risking the generation of erroneous information due to their inability to understand domain-specific vocabulary (Naveed et al., 2023).

This new methodology fills a void by moving beyond simple data extraction to provide combined, narrative-centric insights. Phase one uses a core LLM to create a high-quality, domain-centric dataset, directly addressing the data availability problem (Wu et al., 2025). Phase two then fine-tunes small, lightweight models on this dataset for cost-effective, secure, and rapid on-site deployment (Povoas et al., 2025).

Future Research Directions and Opportunities of Petroleum Engineering Applying Artificial Intelligence

Based on this study, future research will focus on creating a hand-annotated dataset to enable rigorous, metric-based evaluation and establish a benchmark for comparisons. Further work will explore different model sizes to optimize performance, cost, and speed, and will deploy the solution in real operational settings to get real-time feedback from engineers. The long-term vision is to expand the workflow to cover a greater range of unstructured documents, such as safety and geologic reports. Ultimately, this project aims to provide a reproducible methodology for building a scalable AI framework capable of processing all unstructured data across the entire value chain to support wiser strategic decision-making.

Methodology

Dataset and Preprocessing

The methodology for this study is structured around a two-stage AI pipeline designed to convert unstructured legacy well reports into a queryable, structured format. This process begins with data ingestion and preprocessing, followed by synthetic data generation using a large foundational model, and culminates in the fine-tuning of a compact, specialized model for local deployment. The entire workflow is designed for automation, scalability, and practical application in an operational environment.

Dataset and Preprocessing

The foundation of this research is a corpus of 2,587 legacy petroleum well reports sourced from various Egyptian oilfields. The dataset is composed of 2,016 Daily Drilling Reports (DDRs) and 571 Daily Workover Reports (WORs), representing a realistic sample of operational documentation in the industry. These reports were provided in heterogeneous formats, including PDF, Microsoft Word, and Excel, which is typical of legacy data systems.

A robust preprocessing pipeline was developed to handle this diversity. Specialized Python libraries—pdfplumber for PDFs, openpyxl for Excel, and win32com for Word—were employed for initial text extraction. A key aspect of our approach was treating different file formats of the same report (e.g., a PDF and an Excel version) as a form of data augmentation. This technique provided the models with varied textual representations of identical information, enhancing their robustness to formatting idiosyncrasies.

Two-Stage Training Pipeline

Our core technical contribution is a two-stage training pipeline that leverages a large model to create a high-quality dataset for training a smaller, more efficient model. This approach circumvents the costly and time-consuming process of manual data annotation, which is a major bottleneck in developing specialized AI solutions.

Stage 1: Synthetic Data Generation. In the first stage, a foundational large language model, termed the LlamaExtract agent, was used to parse the raw, preprocessed text from the well reports. The agent's task was to extract key information and structure it into a predefined JSON format. This structure was rigorously enforced by a Pydantic schema, which acts as a data validation layer, ensuring that every output conforms to the specified fields, data types, and nested structures. The schema defines the target data model, including fields for well history, operational problems, dates, and summaries as in (Table 2).

Table 2—Json Schema Used for both Building the Dataset and fine-tuning the small model

operation_date	string	The date of the reported operation.
well_name	string	The unique identifier or name of the well.
operation_title	string	A concise, generated title for the day's main activities.
operations	array of objects	A list of distinct operations performed.
operations.operation_type	string	The category of the operation (e.g., "Drilling," "Tripping").
operations.depth_interval	string	The depth range where the operation occurred (e.g., "1500-1600 ft").
operations.description	string	A detailed narrative of the specific operation.
problems_reported	array of objects or null	A list of any issues or non-routine events encountered.
problems_reported.description	string	A description of the problem.
problems_reported.solution	string or null	The solution or mitigation action taken for the problem.
main_content	string	The full, cleaned text of the report's main body in plain english.
main_content_summary	string	An AI-generated summary of the main_content.

For each of the 2,587 reports, the LlamaExtract agent generated a corresponding structured JSON object. This process resulted in a dataset of 2,587 instruction-response pairs, where the "instruction" is the raw report text and the "response" is the structured JSON output. This synthetic dataset, formatted in JSONL (JSON Lines), became the ground truth for the second stage of the pipeline.

Stage 2: Fine-tuning. In the second stage, a compact 7-billion-parameter model, Qwen2.5-7B-Instruct-bnb-4bit, was fine-tuned using the synthetically generated dataset. To make this process computationally efficient, the

Parameter-Efficient Fine-Tuning (PEFT) method, specifically Low-Rank Adaptation (LoRA), was employed. This technique dramatically reduces the number of trainable parameters, enabling the process on a cloud-based GPU. The model was trained on a premium A100 GPU with the specific goal of producing valid JSON output that strictly conforms to a predefined schema. Key techniques included 4-bit quantization to reduce memory usage and a cosine learning rate scheduler over 7 epochs to ensure stable convergence.

Table 3—Model Fine-Tuning Experiment Parameters

Parameter	Value	Rationale
Architecture		
Max Sequence Length	8,192 tokens	Accommodate long reports
Quantization	4-bit	Memory efficiency
Precision	bfloat16	Numerical stability (A100)
LoRA Configuration		
Rank (r)	64	Complex domain knowledge
Alpha	64	Balanced learning rate scaling
Target Modules	q,k,v,o,gate,up,down_proj	Full attention + MLP
Dropout	0.05	Regularization
Training Hyperparameters		
Batch Size	$2 \times 32 = 64$ (effective)	Memory constraints
Learning Rate	0.0001	Conservative adaptation
Scheduler	Cosine	Better convergence
Epochs	7	Sufficient for 2,587 samples
Optimizer	AdamW (fused)	A100 optimization

Evaluation Framework

A dual-pronged evaluation framework was established to assess the model's performance comprehensively, combining automated quantitative metrics with qualitative human-in-the-loop validation.

Quantitative Metrics. A test set of 50 randomly selected reports was used for quantitative evaluation. The framework included:

- **Field-level Accuracy:** For scalar fields like dates or well names, an exact string match was used. For array fields (e.g., a list of operations), the F1-score was calculated to balance precision and recall, as order and exact phrasing can vary.
- **Semantic Similarity:** For free-text fields like summaries, semantic similarity was measured (using cosine similarity on text embeddings) with a threshold of 0.8, to assess contextual correctness rather than penalizing minor wording differences.
- **Weighted Overall Score:** A single, weighted score was computed to represent overall performance, with weights assigned based on the operational importance of each field (e.g., the list of operations was weighted most heavily).
- **JSON Parsing Success Rate:** The percentage of model outputs that were syntactically valid JSON and conformed to the schema was measured.

Qualitative Assessment. The model’s output was reviewed by two petroleum engineers with domain expertise. They assessed the outputs for technical accuracy, completeness, coherence, and operational utility, providing a real-world sanity check that goes beyond statistical measures. This expert validation is crucial for ensuring the model is not only statistically accurate but also practically useful and reliable for decision-making.

The workflow, summarized in [Figure 1](#), includes data preprocessing, schema-guided synthetic data generation, and fine-tuning of the 7B model using PEFT. Final evaluation combines quantitative metrics and expert validation to ensure accuracy and deployment readiness.

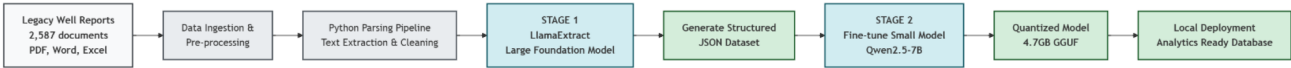


Figure 1—Overview of the data processing, training, and evaluation workflow for structured information extraction from petroleum reports.

Results

The performance of the fine-tuned 7B model was evaluated using the dual framework of quantitative metrics and qualitative expert review. The model was running locally using **Ollama** deployed as a single file (GGUF) and using it for testing the 50 samples. The Results indicate that the model is effective at structured data extraction, demonstrating both technical accuracy and practical utility.

Quantitative Performance

The model demonstrates strong accuracy in extracting structured fields such as operation date and well name, moderate performance on text fields, and lower accuracy on array fields. (Table 4) summarizes the quantitative performance on 50 test samples, with an overall weighted accuracy of 70.4% and a 98% JSON extraction success rate.

Table 4—Quantitative performance metrics on 50 test samples.

Metric Category	Field/Component	Accuracy (%)	Min-Max (%)	Weight	Evaluation Method
Overall Performance	Weighted Average	70.4 ± 10.3	50.0 - 85.6	1	Weighted combination
Scalar Fields	Operation Date	100.0 ± 0.0	100.0 - 100.0	0.15	Exact string match
Scalar Fields	Well Name	98.6 ± 5.3	85.0 - 100.0	0.15	Exact string match
Scalar Fields	Operation Title	83.4 ± 11.6	60.0 - 95.0	0.1	Semantic similarity (≥0.85)
Array Fields	Operations	45.7 ± 17.6	20.0 - 70.0	0.3	F1-score (precision/recall)
Array Fields	Problems Reported	38.2 ± 20.4	15.0 - 65.0	0.1	F1-score (precision/recall)
Text Fields	Main Content	76.4 ± 15.3	50.0 - 95.0	0.1	Semantic similarity (≥0.8)
Text Fields	Main Content Summary	72.1 ± 19.7	45.0 - 95.0	0.1	Semantic similarity (≥0.8)
System Performance	JSON Parsing Success	98	-	-	Valid JSON structure

Qualitative Validation

Domain expert evaluation (Table 5) confirms the model’s technical accuracy, data completeness, and readiness for deployment. It produced well-structured JSON output and required only minor post-processing.

Table 5—Domain expert evaluation (n=2 petroleum engineers) confirming technical accuracy and operational deployment readiness.

Evaluation Criterion	Score	Expert Comments
Technical Accuracy	3.8/5	Correctly identifies petroleum terminology and operational sequences. Minor issues with complex equipment names.
Data Completeness	4.0/5	Captures majority of critical operations. Occasionally misses secondary activities mentioned in narrative text.
Schema Compliance	4.7/5	Excellent JSON structure adherence. Rare formatting inconsistencies in depth intervals.
Operational Utility	4.0/5	Ready for operational deployment with minor post-processing. Significant time savings over manual extraction.

Key Findings

Based on the combined results, the key findings are as follows:

- Reliable extraction of critical operational data
- Robust handling of petroleum industry jargon
- Consistent JSON schema compliance for automated workflows
- Significant efficiency gains over manual processing

Discussion

The results of this study provide strong evidence for the viability of a two-stage AI pipeline for automated data extraction from legacy petroleum reports. The findings have important implications for both the technical implementation of specialized AI and its practical application within the energy sector.

Technical Contributions

This work makes several key technical contributions.

- The **two-stage training pipeline** effectively addresses the manual annotation bottleneck that often hinders the development of domain-specific AI. By using a foundational model to generate a synthetic dataset, we created a scalable and reproducible method for training specialized models.
- The use of **domain-specific fine-tuning** on a compact, 7B-parameter model demonstrates that highly specialized performance can be achieved without relying on massive, proprietary models
- The final **quantized deployment** model is lightweight and enables entirely local inference, which is critical for data privacy, security, and cost-effectiveness, as it avoids dependencies on external APIs.

Performance Analysis

A detailed analysis of the results confirms the effectiveness of our approach. The model's near-perfect accuracy on scalar fields like Operation Date (100%) and Well Name (98.6%) highlights its proficiency at fact-based, low-ambiguity extraction tasks. The moderate-to-high semantic similarity scores for summary and content fields (72-76%) are particularly encouraging, as they indicate the model is not merely matching keywords but is genuinely comprehending and paraphrasing narrative content.

The lower F1-scores for array fields (Operations at 45.7% and Problems Reported at 38.2%) warrant further discussion. This is the most complex task, requiring the model to perform several sub-tasks: identify multiple distinct items in a text, segment them correctly, and summarize each one. The F1-score, which penalizes both false positives and false negatives, reflects this difficulty. While the score indicates room for improvement, the domain expert evaluation confirmed that the model still captured the majority of critical operations, rating data completeness at 4.0/5. This suggests that while the model's output may not be a perfect replica of a human-annotated list, its operational utility is already high.

Furthermore, the 98% JSON parsing success rate is a crucial result for automation. It ensures that the output is consistently structured and can be reliably ingested by downstream analytics platforms and databases, which is a primary objective of the workflow. The combination of a strong LoRA configuration ($r=64$) and a cosine learning scheduler proved effective for domain adaptation, allowing the 7B model to achieve a level of performance that validates our hypothesis about targeted fine-tuning.

Limitations. Despite the promising results, this study has several limitations.

- The evaluation was conducted exclusively on reports from Egyptian oilfields, and the model's performance may vary on documents from other regions with different terminology or formatting conventions.
- The schema is complex and was designed by domain experts; modifying or extending it for different document types may require similar expertise.
- The entire workflow's performance is highly dependent on the quality of the synthetic data generated in Stage 1. Any errors or biases in the foundational model's output will be propagated to the fine-tuned model.
- The generated data need advanced post processing techniques in order to start actually using them for analytics and decision making.

Conclusion

This work presents a scalable AI pipeline for petroleum data extraction, transforming 2,587 legacy reports into structured assets. The two-stage approach (AI Annotation generation + fine-tuning) achieves 70.4 % accuracy while enabling local deployment. Key contributions include domain-specific LoRA optimization and comprehensive evaluation framework.

Future Work

1. Expanded Evaluation: Multi-region dataset validation with diverse oilfield terminology
2. Model Scaling: Comparative analysis across 3B-14B parameter range for optimal deployment
3. Post-Extraction Data Integration: Address well naming inconsistencies across files, temporal data consolidation, and duplicate operation detection through entity resolution algorithms.
4. Strategic Analytics Development: Transform structured data into actionable insights via advanced data analysis techniques for textual data.
5. Real-World Deployment: Operational testing with continuous feedback loops
6. Multi-Document Processing: Expand to geological surveys, completion reports, and regulatory filings

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